Mathematics Bootcamp

Functions, Counting, The Real Numbers

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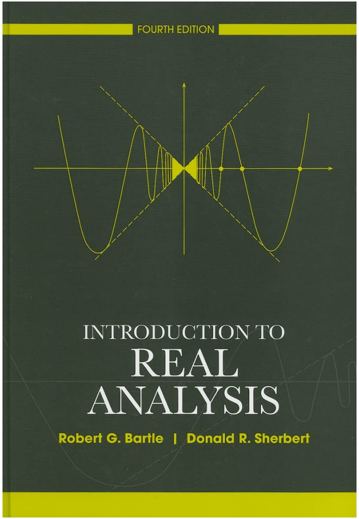
Bartle-Sherbert - Real Analysis



- Most of what follows is taken from Bartle-Sherbert.
- My favorite introductory book for Graduate Level Math.
  - Careful attention to detail.
  - Great pedagogical order.

Cartesian Product

1.1.5 Definition

If A and B are nonempty sets, then the **Cartesian product** $A \times B$ of A and B is the set of all ordered pairs (a, b) with $a \in A$ and $b \in B$. That is,

$$A \times B := \{(a, b) : a \in A,\ b \in B\}.$$

For example, if $A := \{x \in \mathbb{R} : 1 \leq x \leq 2\}$ and $B := \{y \in \mathbb{R} : 0 \leq y \leq 1 \text{ or } 2 \leq y \leq 3\}$, then instead of a rectangle, we should have a drawing such as Figure 1.1.3.

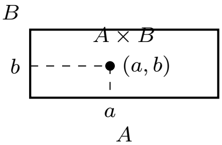


Figure 1.1.2

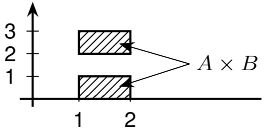


Figure 1.1.3

Functions

1.1.6 Definition

Let A and B be sets. Then a **function** from A to B is a set f of ordered pairs in $A \times B$ such that for each $a \in A$ there exists a unique $b \in B$ with $(a, b) \in f$. (In other words, if $(a, b) \in f$ and $(a, b') \in f$, then $b = b'$.)

The set A of first elements of a function f is called the **domain** of f and is often denoted by $D(f)$. The set of all second elements in f is called the **range** of f and is often denoted by $R(f)$. Note that, although $D(f) = A$, we only have $R(f) \subseteq B$. (See Figure 1.1.4.)

When $A, B \subseteq \mathbb{R}$, uniqueness is the *vertical line test*: the graph meets each line $x = a$ at most once. Since $D(f) = A$, it meets that line exactly once for every $a \in A$.

Functions as Graphs

- A function f maps a to *exactly one* b: $f(a) = b$ or $a \mapsto b$

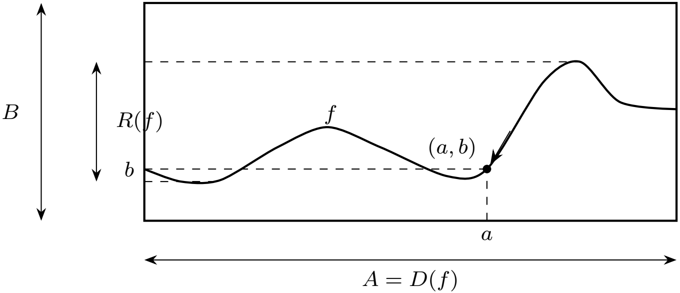


Figure 1.1.4 A function as a graph

Composition and Inverse Images

Composition

Let $f : X \to Y$ and $g : Y \to Z$. Their composition is the function

$$g \circ f : X \to Z, \qquad (g \circ f)(x) := g\big(f(x)\big).$$

The order matters: apply f first and then apply g.

Inverse Image

For a set $D \subseteq Y$, the inverse image of D under f is

$$f^{-1}(D) := \{x \in X : f(x) \in D\}.$$

The notation $f^{-1}(D)$ is defined for every function; f need not be bijective.

Injective, Surjective, and Bijective

1.1.9 Definition

Let $f : A \to B$ be a function from A to B.

(a) The function f is said to be **injective** (or **one-to-one**) if whenever $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$. If f is an injective function, we also say that f is an **injection**.

(b) The function f is said to be **surjective** (or to map A **onto** B) if $f(A) = B$; that is, if the range $R(f) = B$. If f is a surjective function, we also say that f is a **surjection**.

(c) If f is both injective and surjective, then f is said to be **bijective**. If f is bijective, we also say that f is a **bijection**.

Proofs - Injective and Surjective

- In order to prove that a function f is injective, we must establish that:

  $$\text{for all } x_1, x_2 \text{ in } A, \quad \text{if } f(x_1) = f(x_2), \text{ then } x_1 = x_2.$$

  To do this we assume that $f(x_1) = f(x_2)$ and show that $x_1 = x_2$.

- To prove that a function f is surjective, we must show that for any $b \in B$ there exists at least one $x \in A$ such that $f(x) = b$.

Cardinality - Finite and Infinite Sets

1.3.1 Definition

(a) The empty set $\varnothing$ is said to have 0 **elements**.

(b) If $n \in \mathbb{N}$, a set S is said to have n **elements** if there exists a bijection from the set $\mathbb{N}_n := \{1, 2, \ldots, n\}$ onto S.

(c) A set S is said to be **finite** if it is either empty or it has n elements for some $n \in \mathbb{N}$.

(d) A set S is said to be **infinite** if it is not finite.

1.3.6 Definition

(a) A set S is said to be **denumerable** (or **countably infinite**) if there exists a bijection of $\mathbb{N}$ onto S.

(b) A set S is said to be **countable** if it is either finite or denumerable.

(c) A set S is said to be **uncountable** if it is not countable.

Infinite Sets Can Have Different Cardinalities

Comparing Sizes

Two sets have the same cardinality when there is a bijection between them. This definition applies to both finite and infinite sets.

- $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ are countably infinite: each can be put in bijection with $\mathbb{N}$.
- $\mathbb{R}$ is uncountable: there is no bijection $\mathbb{N} \to \mathbb{R}$.

Cantor's diagonal argument proves the second claim: video.

Why the Real Numbers Matter

- Economic alternatives often lie in $\mathbb{R}^L$, while utility is real-valued.
- We need the structure of $\mathbb{R}$, not a construction of it.
- Three ingredients will matter:
  - order, for comparisons;
  - completeness, for limits and extrema;
  - distance, for neighborhoods and convergence.

$\mathbb{R}$ Is a Complete Ordered Field

Ordered Field

A **field** supports addition and multiplication satisfying the field axioms. A **total order** compares every pair of elements and is compatible with the field operations.

$$a < b \implies a + c < b + c,$$
$$0 < a \text{ and } 0 < b \implies 0 < ab.$$

Totality gives **trichotomy**: for any a, b, exactly one of $a < b$, $a = b$, or $a > b$ holds.

The Structural Difference

Both $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. Of these two, only $\mathbb{R}$ is **complete**: every nonempty set of real numbers bounded above has a least upper bound in $\mathbb{R}$.

Absolute Value and the Real Line

2.2.1 Definition

For $a \in \mathbb{R}$, its **absolute value**, denoted $|a|$, is

$$|a| := \begin{cases} a, & \text{if } a > 0, \\ 0, & \text{if } a = 0, \\ -a, & \text{if } a < 0. \end{cases}$$

The Real Line

A convenient and familiar geometric interpretation of the real number system is the real line. In this interpretation, the absolute value $|a|$ of an element a in $\mathbb{R}$ is regarded as the distance from a to the origin 0. More generally, the **distance** between elements a and b in $\mathbb{R}$ is $|a - b|$. (See Figure 2.2.3.)

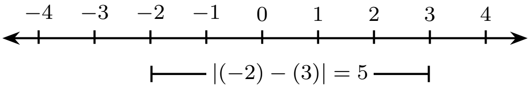


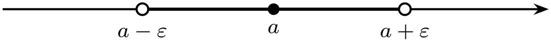
Figure 2.2.3 The distance between $a = -2$ and $b = 3$

Neighborhoods in $\mathbb{R}$ and $\mathbb{R}^L$

2.2.7 Definition

For $a \in \mathbb{R}$ and $\varepsilon > 0$, the ε-**neighborhood** of a is

$$V_\varepsilon(a) := \{x \in \mathbb{R} : |x - a| < \varepsilon\} = (a - \varepsilon, a + \varepsilon).$$



An ε-neighborhood of a

Euclidean Neighborhood

For $x \in \mathbb{R}^L$ and $\varepsilon > 0$,

$$B_\varepsilon(x) := \{y \in \mathbb{R}^L : \|y - x\| < \varepsilon\}, \qquad \|z\| := \sqrt{\textstyle\sum_{\ell=1}^{L} z_\ell^2}.$$

The Triangle Inequality Controls Distances

Triangle Inequality

For every $x, y, z \in \mathbb{R}^L$,

$$\|x - z\| \leq \|x - y\| + \|y - z\|.$$

Geometric Meaning

The direct distance from x to z cannot exceed the distance obtained by travelling from x to z through an intermediate point y.

Sequence Convergence in $\mathbb{R}^L$

Convergence

A sequence (x_n) in $\mathbb{R}^L$ **converges** to $x \in \mathbb{R}^L$ if

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N, \qquad \|x_n - x\| < \varepsilon.$$

We write $x_n \to x$. Equivalently, every neighborhood of x contains all terms of the sequence from some point onward.

Coordinatewise Convergence

For $x_n = (x_{n1}, \ldots, x_{nL})$ and $x = (x_1, \ldots, x_L)$,

$$x_n \to x \quad \Longleftrightarrow \quad x_{n\ell} \to x_\ell \quad \text{for every } \ell = 1, \ldots, L.$$

Continuity of a Function

Sequential Definition

Let $X \subseteq \mathbb{R}^L$, $Y \subseteq \mathbb{R}^K$, and $f : X \to Y$. The function f is *continuous at* $x \in X$ if, for every sequence (x_n) in X,

$$x_n \to x \quad \Longrightarrow \quad f(x_n) \to f(x).$$

It is continuous on X if it is continuous at every $x \in X$.

Keep the Objects Separate

This is continuity of the function f. Day 3 defines continuity of a preference relation through closed upper and lower contour sets.

Bounds

2.3.1 Bounds

Let $\varnothing \neq S \subseteq \mathbb{R}$.

$$\begin{aligned} u \text{ is an upper bound} \quad &\iff \quad s \leq u \quad \text{for every } s \in S, \\ w \text{ is a lower bound} \quad &\iff \quad w \leq s \quad \text{for every } s \in S. \end{aligned}$$

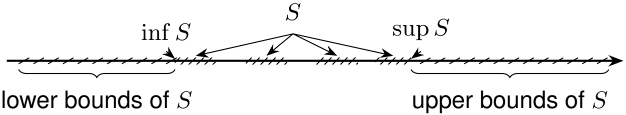
S is **bounded** if it has both an upper and a lower bound.

Example

For $S = (0, 1)$,

$$\text{upper bounds} = [1, \infty), \qquad \text{lower bounds} = (-\infty, 0].$$

Suprema and Infima



2.3.2 Supremum = least upper bound

$$u = \sup S$$

1. $s \leq u$ for every $s \in S$.
2. $u \leq v$ for every upper bound v.

$$u \in S \Rightarrow u = \max S;$$
$u \notin S \Rightarrow$ no maximum.

Infimum = greatest lower bound

$$w = \inf S$$

1. $w \leq s$ for every $s \in S$.
2. $t \leq w$ for every lower bound t.

$$w \in S \Rightarrow w = \min S;$$
$w \notin S \Rightarrow$ no minimum.

Completeness of the Real Numbers

2.3.6 Completeness Property of $\mathbb{R}$

For every nonempty $S \subseteq \mathbb{R}$,

$$S \text{ bounded above} \quad \Longrightarrow \quad \sup S \in \mathbb{R}.$$

Consequently, if S is bounded below, then $\inf S \in \mathbb{R}$.

Why $\mathbb{Q}$ Is Not Complete

$$A = \{q \in \mathbb{Q} : q^2 < 2\}$$

is bounded above, but its supremum in $\mathbb{R}$ is $\sqrt{2} \notin \mathbb{Q}$. Thus A has no supremum in $\mathbb{Q}$.

Completeness means that $\mathbb{R}$ has no gaps.

Key Implications for Proofs

Archimedean Property

2.4.3 *If $x \in \mathbb{R}$, then there exists $n_x \in \mathbb{N}$ such that $x \leq n_x$.*

2.4.5 Corollary *For every $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $1/n < \varepsilon$.*

Density Results

2.4.8 Density Theorem *If x and y are any real numbers with $x < y$, then there exists a rational number $r \in \mathbb{Q}$ such that $x < r < y$.*

2.4.9 Corollary *If x and y are real numbers with $x < y$, then there exists an irrational number z such that $x < z < y$.*

Utility Is Ordinal

Ordinal Utility

A function $u : X \to \mathbb{R}$ represents $\succeq$ if

$$x \succeq y \quad \Longleftrightarrow \quad u(x) \geq u(y).$$

If $\phi : \mathbb{R} \to \mathbb{R}$ is strictly increasing, then $\phi \circ u$ represents the same preferences. Utility levels and differences have no intrinsic meaning.

Existence

Every utility representation induces rational preferences. On finite or countable X, rationality is also sufficient. On the standard Euclidean choice set $X = \mathbb{R}^L_+$, continuity is sufficient for a continuous representation; Day 3 states the theorem.

Back to Economics - Utility Functions

Utility Functions

In economics, we often describe preference relations by means of a *utility function*. A utility function $u(x)$ assigns a numerical value to each element in X, ranking the elements of X in accordance with the individual's preferences. This is stated more precisely in Definition 1.B.2.

Definition 1.B.2: A function $u: X \to \mathbb{R}$ is a *utility function representing preference relation* $\succsim$ if, for all $x, y \in X$,

$$x \succsim y \Leftrightarrow u(x) \geq u(y).$$

Utility Representation Implies Rationality

The ability to represent preferences by a utility function is closely linked to the assumption of rationality. In particular, we have the result shown in Proposition 1.B.2.

Proposition 1.B.2: A preference relation $\succsim$ can be represented by a utility function only if it is rational.

MWG Exercises: Utility Representation

1.B.4[A] Consider a rational preference relation $\succsim$. Show that if $u(x) = u(y)$ implies $x \sim y$ and if $u(x) > u(y)$ implies $x \succ y$, then $u(\cdot)$ is a utility function representing $\succsim$.

1.B.5[B] Show that if X is finite and $\succsim$ is a rational preference relation on X, then there is a utility function $u: X \to \mathbb{R}$ that represents $\succsim$. [*Hint*: Consider first the case in which the individual's ranking between any two elements of X is strict (i.e., there is never any indifference), and construct a utility function representing these preferences; then extend your argument to the general case.]